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Concussion & Head-Related Injury in Sports Prevalence

Problem Statement:

Head-related injuries and concussions can have serious effects on the individual, such as differences in behavior, mood, memory, and emotions. To effectively prevent and target health education on head-related injuries, it is important to understand where head-related injuries are most prevalent in terms of demographics and type of sport.

Data Sources:

- Physical Activity from Youth Risk Behavior Surveillance System (High School)
- Rates of TBI-related ED Visits, Hospitalizations and Death by Sex - United States, 2001 - 2010
- Games and Practices with Associated Injury Rates (NCAA ISS)

Goals of Analysis:

- To find whether sports-related head injuries are prevalent within at least 10% of the population in the United States
- To visualize trends in differences in prevalence between demographics such as gender and age to identify what groups have been most impacted by sports-related head injuries
- To look at what sports have the highest risk for head-related injury

Testable Hypotheses:

1. Sports-related head injuries occur in at least 10% of the U.S. population.
2. Males have a higher prevalence of head injuries in sports than females.
3. Contact sports (e.g., football, soccer) are associated with a higher risk of concussions compared to non-contact sports (e.g., swimming, golf).

Algorithms to use:

Regression Analysis: To assess relationships between demographic factors (age, gender) and the likelihood of head injuries.

Data Visualization: To present trends and differences in injury rates across different groups.

Ethical Considerations:

- **Privacy:** The datasets contain sensitive health information, especially in the data about traumatic brain injuries (TBI) and surveys of young people. It's important to make sure that the identity of individuals remains private and that their personal information is kept confidential.
- **Bias**: There could be biases in data collection, such as underreporting of injuries among certain demographics (e.g., females in traditionally male-dominated sports). Care must be taken to ensure analysis accounts for potential biases and does not misrepresent the true scope of the problem.

Division of Labor:

- Adam:
- Background in Health Science, will look at dataset for TBI-related ED Visits
- Create data visualizations to understand hospitalization differences by different demographics
- Nadim:
- Background in Finance, will help analyze data trends
- Assist in setting up regression analyses to test hypotheses and collaborate with Adam to create clear, impactful visuals that communicate risk trends
- Henri:
- Background in economics, will help find accurate datasets to test our hypothesis
- Will help create regressions to create visualizations for our project
- Jack:
- Background in Health Science, will help identify factors which increase likelihood of sports-related head injuries
- Will create a ML model that can predict likelihood that someone will/or has attained a sports-related injury based on their history and activity